QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 7

RULE · Vertex form (cho'qqi shakli)

$$y = a(x-h)^2 + k$$

Vertex $= (h, k)$, axis of symmetry $x = h$. $a > 0$ opens up, $a < 0$ opens down; larger $|a|$ = narrower.

$y = 3(x-1)^2 + 4 \implies$ vertex $(1, 4)$, axis $x = 1$, opens up

RULE · x-intercepts = roots

Set $y = 0$ and solve. (y-intercept is different – the value at $x = 0$, equal to c.)

$y = x^2 - 7x + 10 \implies (x-2)(x-5) = 0 \implies x = 2$ or 5

"x-intercept," "root," and "solution" all mean the same thing.

RULES SHEET – 2/2

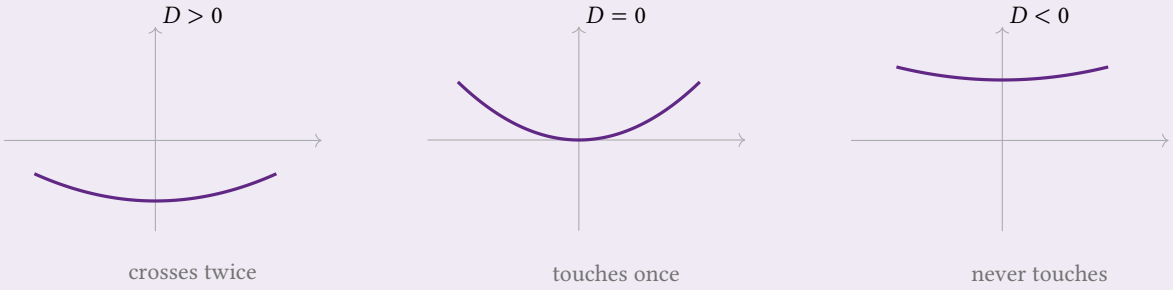
RULE · Discriminant (diskriminant)

$$D = b^2 - 4ac$$

$D > 0$: 2 real roots $D = 0$: 1 real root $D < 0$: 0 real roots

$x^2 + 2x + 5 = 0 \implies D = 2^2 - 4(1)(5) = -16 \implies$ no real roots

RULE · What each case looks like



crosses twice touches once never touches